

Designing a Web App to Visualize 3D Blood Flow Patterns in Cerebral Aneurysms

Abstract

Problem: Approximately, 1 in 50 people in the United States suffer from cerebral aneurysms. Of these, 1-2% of aneurysms rupture, resulting in brain hemorrhage and patient death. Since neurological treatment approaches are complex and subject the patient to significant risk, there is a need for selective treatment based on aneurysm severity. Currently, clinical diagnosis using an angiogram yields a black-and-white image of the hemodynamic changes in the artery which is often inconclusive of the severity of the aneurysm. Therefore, there is a need for a more effective method for visualization and quantification of angiographic data towards accurate diagnosis of cerebral aneurysms.

Methods: Simulated angiographic data in .csv and STL format was provided as input for the application. Rigorous coding using Python 3.11 and associated libraries (e.g., matplotlib, numpy) was performed to process the input files and merge them into an interactive 3D environment that displayed vector magnitude and wall shear stress information.

Results: An open-source flask-based web application was created to serve as a user interface that allowed for uploading of data input files and displaying of quantitative vector and wall shear stress plots. These plots were categorized by a quantitative spectrum of colors based on vector magnitude and degree of wall shear stress for easier visualization. This application allowed for data manipulation in 3D to better highlight specific regions of the artery that may be of concern to clinicians.

Conclusions: The created open-source web application allowed for large-scale access to better visualization technologies. Hemodynamic observation in this way can also improve diagnosis and post-treatment observation of other cardiovascular diseases such as stenosis, other aneurysms, and strokes.

Introduction

A cerebral aneurysm (also known as intracranial aneurysm) is a bulge or ballooning in a blood vessel in the brain (Mayo Foundation for Medical Education and Research, 2023). An estimated 1 in 50 people in the US are impacted by cerebral aneurysms, and the majority of them are undiagnosed throughout their lifetime (Brain Aneurysm Foundation, 2023). In addition, less than 1% of cerebral aneurysms rupture every year (Jeong & Rhee, 2012). Due to the large degree of uncertainty surrounding the rupturing and severity of cerebral aneurysms, accurate diagnosis is key in order to minimize risk, while avoiding unnecessary treatment. This issue is further heightened due to the treatment procedure for cerebral aneurysms entailing either neurological surgery (known to have varying success rates) or non-invasive procedures such as endovascular therapy that can fail in up to 40% of cases (White et al, 2023).

Based on this information, it is clear that appropriate diagnosis of cerebral aneurysms is crucial in creating a successful treatment plan. The most widely used diagnosis method for cerebral aneurysms is angiographic injection, a process which utilizes a contrast medium coupled with an angiogram to view the hemodynamics (blood flow) in an artery (Moore, 2017). Further study of such hemodynamics through Computational Fluid Dynamics (CFD) can better diagnose cerebral aneurysms and their need for treatment. However, a cerebral angiogram (CA) is difficult to interpret for clinicians as the data is largely qualitative in nature. Studies have developed quantitative parameters for easier diagnosis (White et al, 2023), but such procedures require significant computational power and resources that most clinicians lack. As a result, CA data needs to be converted into a form that allows for easier visualization. These can include Wall Shear Stress (WSS) plots, contour plots, and vector magnitude plots. Such methods have been visualized using specialized CFD software which can have high upfront costs and a steep

learning curve, creating a lack of accessibility for clinicians. Therefore, there is a need for the development of cost-effective user-friendly visualization software to be used by clinicians for better diagnosis of cerebral aneurysms.

Background

Diagnosis:

Cerebral aneurysms are known to impact those around the age of 50, with a larger connection to males over females. An estimated 6.7 million people in the US have an unruptured cerebral aneurysm, but most are undiagnosed due to the lack of symptoms and the complex diagnosis methods of the disease (Brain Aneurysm Foundation, 2023).

The diagnosis of cardiovascular disease using contrast-based scanning methods dates back to 1998 and has been administered over 200 million times (Chandra & Mohan, 2015). The basic procedure for a contrast-based MRI/CT scan is to inject a contrast medium (specific to use case) into the bloodstream and observe the hemodynamic change over the cardiac cycle (White et al, 2023). This process then creates a video animation of the contrast fluid flowing through the targeted artery which can reveal hemodynamic information. This has been effective, but as said before, the issue that remains is the relevant interpretation of the contrast-based MRI or CT scan data through better visualization strategies to diagnose aneurysm severity (White et al, 2023).

Furthermore, studies show that hemodynamic influences are also present in the formation of cerebral aneurysms (Mantha et al., 2006). In other words, hemodynamic changes can show the likelihood of an aneurysm forming in the future as well as the existing methods of diagnosis. This creates a preventive care application for hemodynamic analysis, especially in patients with genetic predisposition to the disease.

Issues in Treatment:

Furthermore, treatment of cardiovascular diseases can often be temporary, so there is potential for recurrent rupture of an artery after treatment (Jeong & Rhee, 2012). In cases of aortic stenosis, despite the seemingly limitless durability of a stent (metal tube used to expand a

constricted artery), there is a 2-3% failure rate of the treatment after the fact (Srakocic, 2022). This entails the narrowing of the artery once again, without resolving the underlying issue. The recurrent narrowing of the artery can cause significant damage and inhibit blood flow. This is another application where observation of hemodynamics through 4D Magnetic Resonance Angiography (Bustamante et al., 2017) can outline potential for future issues. These Magnetic Resonance procedures display a video of the cardiac cycle for clinicians to observe hemodynamic change. In this example, routine scans would illustrate the progression of the treatment and limit the chance for recurrent rupture. Action can be taken if the findings observed in the hemodynamics indicate a serious issue (high velocity hemodynamics where an artery constricts). This can decrease the probability and dangers of recurrent rupture in the artery.

There is also significant potential for misdiagnosis (both false positives and negatives) in cerebral aneurysms and other such diseases (Mayer et al., 1996). Clinicians utilizing symptomatic diagnosis for cardiovascular diseases have serious potential to be incorrect in their diagnosis and by association, their treatment plan. This can result in unnecessary treatment which can be dangerous and expensive, or the worsening of the disease in cases of a false negative. This can be solved with better visualization technologies.

Other Applications:

The issues outlined above are also present in similar cardiovascular diseases such as aortic stenosis which is the narrowing of the aorta artery in the heart (Marquis-Gravel. et al., 2016). Specifically, diagnosis of aortic stenosis and its severity is difficult and requires greater visualization technology (Saikrishnan et al., 2014). There are various treatment courses for aortic stenosis such as lipid-lowering therapy, antihypertensive drugs, and anticalcific therapy. The use of a stent is necessary in extreme cases. Due to the wide range of treatment methods, appropriate

diagnosis of the severity of the disease is imperative to select an appropriate treatment plan suited to the patient. Lack of proper diagnosis can have detrimental consequences and greatly increase the fatality rate in cardiovascular diseases due to the aforementioned treatment failure rates and misdiagnosis.

In addition, hemodynamics in an artery can visibly change over the progression of a cardiovascular disease which can be a diagnostic tool (Mantha et al., 2006). This change can significantly impact the Wall Shear Stress (WSS) which is the force put on the wall of the artery by the fluid inside (White et al., 2023). Increase of the WSS beyond a threshold can result in possible rupture of the artery which causes hemorrhage (internal bleeding). To avoid this, hemodynamic observation is important to prevent possible rupture (White et al., 2023).

In other studies, 3D vector environments have been created for purposes such as geosensors (Afghantoloe et al., 2014) and scan-to-BIM models (Bassier et al, 2016). Geosensors are tiny devices that collect data regarding their environment. They are often used to collect information such as motion, temperature, humidity, pollutants, and traffic flow in an area. This information can be displayed in a 3D vector environment which is similar to that required for cardiovascular visualization (Afghantolee et al., 2014). Building Information Modeling (BIM) is the creation of 3D models to represent buildings. This is also done in a 3D vector environment. These studies highlight an alternative use case of vector technologies which can accelerate development in this field.

Software & Purpose:

The limited access of professional-grade software to everyday clinicians presents a need for an effective 3D modeling software that is both easy-to-use and cost-effective. Such software can also be developed for the purpose of medical education where there are clear budget constraints

and in patient information so that patients have a clear understanding of the issues they face. The obvious use cases in third-world countries can also be observed as a no-cost web app will greatly improve healthcare in those regions.

Computational fluid dynamics (CFD) entails the visualization of fluid vectors in 3D and the analysis of that image (White et al., 2023). This can be applied to hemodynamics in various applications such as preemptive care, diagnosis, and post-treatment monitoring. The most prevalent application is to create 3D images of WSS and vector magnitude to determine the severity of cardiovascular diseases.

Specifically, the development of specific functions in the program is necessary. The first of which is a function to visualize the artery through its vector magnitude or WSS. This entails the creation of vector points overlaid on top of a basic structure of the artery and was accomplished by coloring the vector points based on their magnitude and including an STL file (a commonly used CAD format) in the plot. The WSS can be seen through plotting the data with vector points of increased size to show a contour plot. Secondly, the ability to view the artery in 3D is important so that the point of concern can be observed easily. This requires the ability to move, zoom, and turn the image in 3D.

As a backend of this generation, the ability to input files is also important. To make the software accessible to the widest audience, including the capability to input files of various file types is also important. For example, there are significant differences between the loading of a .CSV file, .TXT file, and files with no extension. Rather than having selection input forms, the program needs to have the functionality to automatically detect the file type and load the image accordingly.

Another important feature is ease-of-use as many clinicians lack computational knowledge, and a steep learning curve should not restrict this software from potential users. Based on the literature, the issue in many modern medical softwares lies with the steep learning curve. In addition to this, the affordability of such technologies is also important. The purpose of the software is to reach the widest audience possible, so a free model is significant.

With that in mind, it is established that there is a need for an easy-to-use cost-effective application to improve visualization in cerebral aneurysms and similar cardiovascular diseases.

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Purpose

Cerebral aneurysms are major health complications that affect people between the ages of 30 and 60, impacting 5% of the United States population. Although most aneurysms are not serious and lack symptoms, a ruptured aneurysm can quickly become life-threatening. For this reason, accurate diagnosis of the severity of cerebral aneurysms is critical for developing an effective treatment plan and preventing rupture. The current gold standard for cerebral aneurysm diagnosis entails the use of an angiographic injection which generates a low-resolution black-and-white image of the arterial wall along with minimal hemodynamic information. The lack of information provided by such qualitative diagnostic methods put a great degree of analytical pressure on clinicians, and cloud diagnosis of rupture probability.

In order to combat this problem, better visualization methods that provide quantitative hemodynamic information are required. Computational Fluid Dynamics (CFD) is a viable tool that employs fluid flow principles to create vector magnitude and wall shear stress plots in a 3D environment to accurately depict quantitative hemodynamic information that can aid in reliably diagnosing the severity of cerebral aneurysms. However, existing CFD software requires significantly high computational power and is associated with a steep learning curve. Due to these challenges, clinical use of CFD is rather limited. Therefore, there is a need for an accessible and easy-to-use CFD platform that can be seamlessly adopted by clinicians for more accurate diagnosis of cerebral aneurysm severity.

Engineering Goals

- Development of a desktop app with ability to 3D Vector Plot Models using Python 3
- Development of a web app with the ability to 3D Vector Plot Models using Python 3
- Creation of a Graphical User Interface with the ability to modify the 3D Vector Plot Models

Criteria

- The web app must be able to change color, scale, and other parameters of points
- The web app must be able to accept data from a spreadsheet to create the 3D vector plot
- The program must be written using Python 3
- The program must provide a successful output at least 95% of the time

Materials

- Computer
- Visual Studio Code (interpreter)
- Python 3.11 and Associated Libraries: flask, numpy_stl, matplotlib, mpl_toolkits, plotly, numpy, base64, os, moviepy, pandas, math, io, matplotlib.backends.backend_agg, json

Procedure

1. Create a Python program to generate a customizable 3D model through a desktop app created using the Tkinter library in the development stage
2. Test the program over 50 trials using a variety of data sets (normal, aneurysm, and stenosis), and modify code so that success rate is greater than 95%
3. After achieving the desired accuracy (95%) in the desktop app, create a program to generate a 3D model using a web app
4. Test the new program over 50 trials, and modify code if success rate is less than 95%
5. Deploy the web app on a WSGI server to allow access to the greater Internet

Results

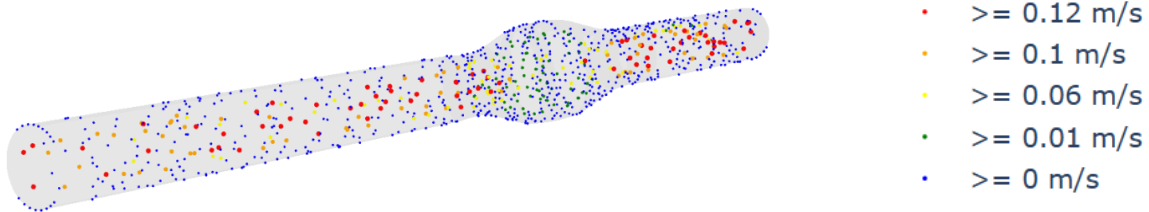
In the development stage, a desktop app using Tkinter was made to outline the potential of matplotlib and related graphing libraries, while hosting the program on a local machine. This was transitioned onto a local web development environment, and then moved to a remote server with access to the greater Internet.

The final deliverable of the study was the creation of an interactive flask-based web application hosted by a WSGI (Web Server Gateway Interface) server. The application allowed for user input of .csv and .txt files and the subsequent generation of a 3D environment using plotly 5.18.0 and the built-in json Python library. The application also had the capability to create a video animation using files of various time points with matplotlib 3.8.2 and moviepy. The final key feature of the application was the generation of Wall Shear Stress (WSS) plots to highlight the risk of rupture in specific areas of the artery.

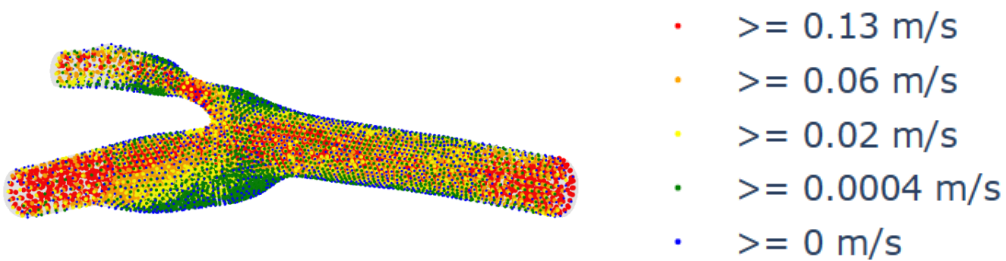
The creation of this web app entailed the development of 3D vector, scatter, and contour plots to represent the vector magnitude or WSS at areas within an artery. Points were separated into various groups based on their respective 3D vector magnitude/WSS and distinguished using an array of colors (red, orange, yellow, green, blue, respectively), quantifying aneurysm severity.

In the generated images, the magnitude difference between the inner and outer parts of the artery can be seen (laminar flow), as well as the disruption caused by cardiovascular disease (aneurysm or stenosis).

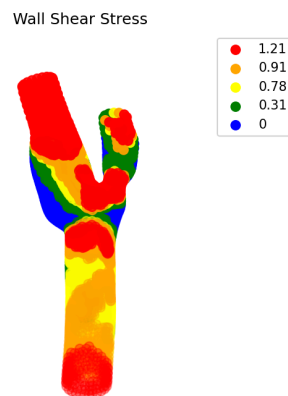
Images (simulated data)



The hemodynamics of a simple aneurysm



The hemodynamics of a stenosis case in the carotid artery



The Wall Shear Stress of a stenosis case in the carotid artery (higher stress = greater chance of rupturing)

Conclusions

All three of the stated engineering goals were met. Firstly, in the development stage, the Tkinter desktop application allowed for generation of vector plots of hemodynamic data locally on a specific machine. Secondly, the outcomes of the desktop application served as a basis to build a flask-based web application that can be remotely accessed to generate vector and WSS plots of hemodynamic data anywhere in the world. Finally, the use of Plotly and Matplotlib in the web application and desktop application respectively created a 3D environment for easier visualization of vector magnitude data.

Furthermore, the four stipulated criteria for the final application were also met. The web application allowed for changing of color, scale, and other parameters to deliver an interactive final image for quantitative hemodynamic analysis. Data was accepted from .csv or .txt files and the Python 3 program resulted in an accurate output in more than 95% of the test cases.

Overall, the web application developed in this project is a low cost user-friendly platform that has the potential to significantly transform medical imaging for cardiovascular diseases such as cerebral aneurysms and carotid stenosis. The availability of this application on the greater Internet will enable widespread adoption of CFD in the clinical arena. The use of a 3D environment to display cardiovascular disease images can allow for quantitative analysis of the severity of the disease towards prevention of arterial rupture.

Future Studies

In future work, the web application can be further developed by integrating artificial intelligence and machine learning algorithms to enable automated analysis of patient hemodynamic data. Further, this advancement coupled with other diagnostic parameters will improve the quantitative landscape and allow for more accurate prediction of disease progression and arterial rupture probability. Together, these developments will allow clinicians to make more informed decisions on an effective treatment plan to reduce mortality associated with cardiovascular diseases.

Another key area for future development is the creation of an augmented reality visualization platform for real-world immersion in 3D. This technology has large implications in patient information, disease progression tracking, and in assistive technology in a surgical environment.

Finally, development of a mobile application would have the potential to aid hemodynamic and laminar flow education to the next generation of students across the globe.